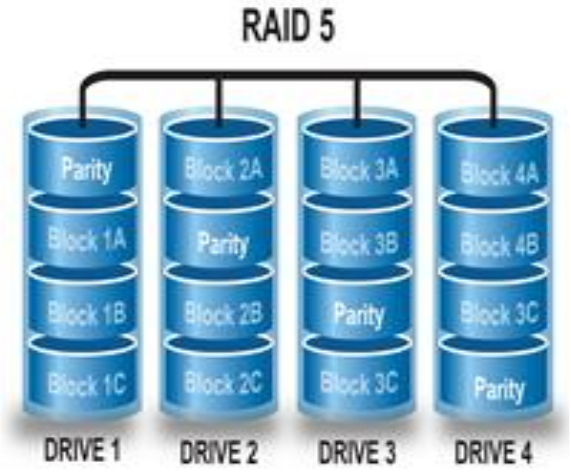


CLOUD COMPUTING COURSE

Lecture21- Cloud Storage (4)

ABSTRACT
Lecture21 Introduces RAID technology.

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Centralized Storage

RAID (Redundant Array of Inexpensive Disks or Drives, or Redundant Array of Independent Disks): A data storage virtualization technology that combines multiple physical disk drive components into one or more logical units for the purposes of data reliability, performance improvement, or both.

Benefits of RAID

RAID allows us to combine drives, increase speed and improve reliability so we can store more data while our data is stored safer.

RAID allows us to bundle multiple hard-drives that your Operating System (Windows, MacOS X, Linux, etc) will see as one single big drive.

Example

As an example, based on Windows and 3 hard-drives of say 1 Tb: Without RAID, taking Windows as an example, 3 drives could be seen as the drives C:, D:, and E: of each 1Tb (if each drive would have one partition). With RAID, these 3 drives can be combined to a single hard-drive, C:, with a capacity of 3Tb (depending on the RAID configuration you choose).

Where can you find RAID?

RAID can be found in any computer, server, external disk array or NAS (Network Attached Storage), where the “controller” can actually be hardware or software.

A RAID controller is a hardware device or software program used to manage hard disk drives (HDDs) or solid-state drives (SSDs) in a computer or storage array so they work as a logical unit. A RAID controller provides a degree of protection for stored data and may also help to improve computing performance by accelerating access to stored data.

Certain SSD manufacturer, to make very fast SSD drives even faster, actually utilize RAID in their SSD controllers, where they combine two (or more) SSD “drives” into one very fast SSD drive.
RAID is available for most hard-drive interfaces like SCSI, IDE/PATA and SATA variations.

RAID Controllers Types

1. Hardware RAID controllers

Hardware RAID controllers are either fixed on the motherboard of a computer, or an add-on card (PCI, PCI Express, etc) or found in an external RAID array solution.

Note that not all motherboards, external RAID arrays and even RAID controllers that claim RAID support actually have RAID support implemented as hardware. Especially the more affordable models often implement RAID in software.

Hardware RAID is the preferred RAID implementation, as it performs faster and seems more reliable, but it comes at a higher price.

Some hardware RAID solutions have a battery backup to protect your RAID during power failure.



RAID Controller card (SATA example)



RAID on a motherboard(Adaptec SATA)

2. Software RAID Controllers

As a cheaper alternative to hardware RAID are RAID solutions implemented in software. This can be a part of the BIOS, firmware on a cheaper RAID controller card, or a driver in the Operating System (no special hardware needed).

The downside of RAID in software is that it performs slower than hardware implementation, although modern solutions perform pretty well, and possibly less reliable compared to a hardware solution.

RAID Configurations (Types)

- RAID 0 – Striping
- RAID 1 – Mirroring
- RAID 2 – Bit level Striping
- RAID 3 / RAID 4 – Striping with Parity
- RAID 5 – Striping with rotating parity
- RAID 6 – Striping with double rotating parity
- Nested RAID (RAID 10)
- JBOD – Just a Bunch of Disks (this is not RAID)

JBOD (not RAID)

JBOD, or Just a Bunch Of Disks, is **not a RAID variation**, but most RAID controllers support it.

With JBOD, hard disks are stringed in a sequence, offering the sum of the capacities of all hard disks in the string.

When writing data to the “array”, first disk 1 will be filled and when full, disk 2 will be filled, etc.

There is no speed gain here, and when a drive fails in a JBOD, not all data is lost, only the data on the defective drive.

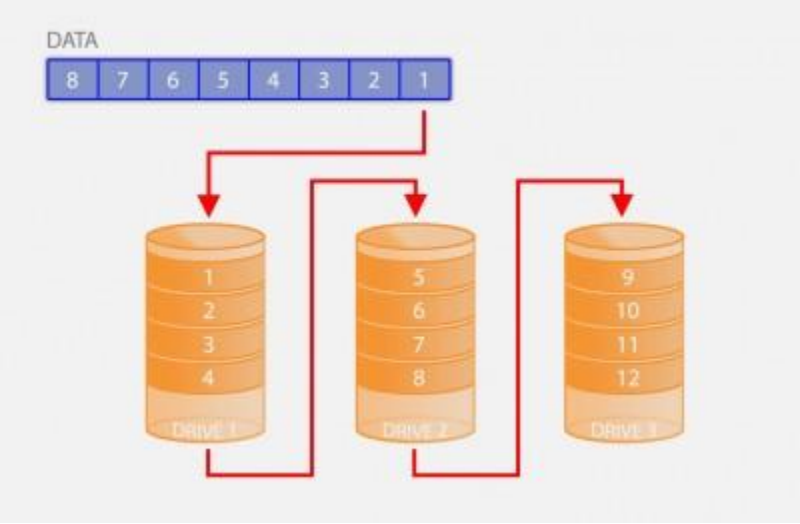
For JBOD disks do NOT need to be of the same size.

Positives:

Simple and **Cheap** way of bundling different sized disks.

Negatives:

No increase in speed or reliability.



JBOD – Just a Bunch Of Disks (not RAID!)